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A Comprehensive Assessment of UPFC-Based Power Flow Control for Voltage Stability Enhancement in Large-Scale Power Systems

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Abstract

This study presents a comprehensive investigation into the optimal deployment of Unified Power Flow Controllers (UPFCs) to enhance voltage stability and reduce power losses in the Sudanese national grid. With the increasing demand for electricity driven by population growth, urban expansion, and industrial development, modern power systems require advanced control strategies to ensure reliable and efficient operation. In this work, the Line Stability Index (Lmn) is employed as a key indicator to identify the most critical transmission lines prone to voltage instability. Based on this index, optimal locations for UPFC installation are determined. Furthermore, an Optimal Power Flow (OPF) framework is utilized to calculate the control parameters of the UPFC devices, aiming to minimize system losses while maintaining operational constraints. The proposed methodology is validated using a real large-scale network model of the Sudanese power system implemented in MATLAB (24b) and NEPLAN (v10) environments. The results demonstrate that installing seven UPFC devices leads to a significant improvement in voltage profiles, maintaining all bus voltages within ±5% of nominal values. Additionally, the system experiences a reduction in total active and reactive power losses by 6.96% and 0.74%, respectively. These findings highlight the effectiveness of UPFC-based control strategies in improving system stability, enhancing transmission efficiency, and supporting the integration of future energy resources.

Keywords: Flexible AC Transmission Systems (FACTS); unified power flow controller (UPFC); voltage stability; Line Stability Index (Lmn); optimal power flow (OPF); power loss reduction

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1. Introduction

The continuous growth in global electricity demand has imposed significant challenges on modern power systems, particularly in developing countries where infrastructure expansion often struggles to keep pace with increasing consumption. In this context, maintaining system stability, improving voltage profiles, and minimizing transmission losses have become critical objectives for power system operators. The Sudanese national grid represents a practical example of such challenges, where rapid population growth, urbanization, and industrial expansion have led to increased stress on the existing transmission network. Traditionally, power system stability has been addressed through network reinforcement, such as constructing new transmission lines or upgrading existing infrastructure. However, these approaches are often associated with high costs, long implementation times, and environmental constraints. Consequently, there has been a growing interest in advanced control technologies that can enhance system performance without requiring extensive physical expansion [1]. In this study, the term existing grid infrastructure refers to the current physical and operational components of the Sudanese power system, including transmission lines, substations, transformers, generation units, and interconnections operating at voltage levels of 500 kV, 220 kV, 110 kV, and 66 kV. These infrastructures are characterized by limited expansion capability, aging equipment, and increasing loading conditions, which necessitate the use of advanced control technologies such as FACTS devices instead of costly network reinforcements.

Flexible AC Transmission System (FACTS) devices have emerged as a powerful and effective solution to address the growing challenges in modern power systems. By leveraging advanced power electronic technologies, FACTS devices enable dynamic control of key transmission parameters, including voltage magnitude, line impedance, and phase angle. This functionality enhances power flow management, alleviates network congestion, and improves overall voltage stability across the grid.

Among the various FACTS technologies, the Unified Power Flow Controller (UPFC) is widely regarded as one of the most versatile and efficient devices. It integrates both series and shunt compensation within a single system, allowing simultaneous control of active and reactive power flows as well as bus voltages. This multifunctional capability makes the UPFC particularly well-suited for addressing complex operational challenges in large-scale power networks.

Beyond performance enhancement, UPFC technology plays a vital role in facilitating the integration of renewable energy sources. As contemporary power systems increasingly depend on intermittent generation such as solar and wind energy, the demand for flexible and adaptive control mechanisms becomes more critical. The UPFC effectively mitigates power fluctuations and helps maintain system stability under varying operating conditions [2].

Despite its numerous advantages, the high installation cost of UPFC devices necessitates careful planning regarding their placement and parameter configuration. Inappropriate allocation may result in suboptimal performance or even negatively impact system stability. Consequently, determining the optimal locations for UPFC deployment remains a significant research challenge. To address this, several voltage stability indices have been introduced in the literature. Among them, the Line Stability Index (Lmn) has demonstrated strong effectiveness in identifying weak transmission lines under heavily loaded conditions. This index provides a quantitative measure of proximity to voltage collapse, making it a valuable tool for guiding the optimal placement of FACTS devices [3].

In this study, a systematic approach is proposed to enhance the performance of the Sudanese national grid through optimal UPFC deployment. The methodology integrates the Line Stability Index (Lmn) for identifying critical lines with an Optimal Power Flow (OPF) framework for determining the optimal control parameters of the devices. The objective is to minimize system losses while maintaining voltage levels within acceptable limits. The proposed approach is implemented and tested on a real large-scale model of the Sudanese electrical network using MATLAB and NEPLAN simulation platforms. The simulations were performed using NEPLAN software for power system analysis and Optimal Power Flow calculations, while MATLAB (MATPOWER toolbox) was used for Lmn index computation and validation of load flow conditions. The results demonstrate the effectiveness of the method in improving voltage stability, reducing transmission losses, and enhancing overall system reliability [4].

This work contributes to the growing body of research on power system optimization by providing a practical and scalable solution for integrating advanced control technologies into existing grid infrastructures. Moreover, it offers valuable insights for decision-makers and system planners aiming to modernize power systems in developing regions [5]. Despite the extensive research on UPFC placement and OPF-based control strategies, most existing studies focus on standard IEEE test systems or simplified network models, with limited application to real large-scale power systems under practical constraints. In addition, the combined use of the Lmn index for location selection and OPF for coordinated parameter tuning has not been sufficiently explored in complex national grids. Therefore, the main contribution of this paper is the development of an integrated and practical framework that combines Lmn-based ranking with OPF-based control for optimal UPFC deployment in a real, large-scale power system. The proposed approach is validated on the Sudanese national grid, providing realistic insights into system performance improvement, voltage stability enhancement, and loss reduction under actual operating conditions. Unlike conventional approaches that rely on either sensitivity indices or optimization techniques independently, the proposed method integrates both in a coordinated framework applied to a real large-scale national grid.

2. Related Work

The application of power electronics in power systems has undergone significant evolution since the introduction of semiconductor devices such as thyristors in the 1970s. Early advancements primarily focused on high-voltage direct current (HVDC) transmission systems and reactive power compensation technologies, including Static VAR Compensators (SVCs) [6].

With continuous progress in semiconductor technologies—particularly the development of Insulated Gate Bipolar Transistors (IGBTs) and Gate Turn-Off Thyristors (GTOs)—more advanced and flexible FACTS devices have been introduced. Among these, the Unified Power Flow Controller (UPFC) has attracted considerable attention due to its capability to simultaneously regulate multiple transmission parameters.

Previous research has investigated various approaches to enhancing voltage stability, including power loss minimization, power factor correction, and optimal generation dispatch strategies. The incorporation of FACTS devices into power system models has significantly improved the accuracy of system analysis under diverse operating conditions [7].

Numerous methodologies have been proposed to determine the optimal placement of FACTS devices within power networks. Many of these approaches rely on voltage stability indices such as the L-index, Fast Voltage Stability Index (FVSI), and Line Stability Index (Lmn). Among these, the Lmn index has demonstrated high sensitivity to variations in reactive power flow, making it particularly effective in identifying weak transmission lines.

In addition, Optimal Power Flow (OPF) techniques have been extensively applied to determine optimal system operating conditions. These methods typically aim to minimize an objective function—most commonly power losses—while ensuring that all system constraints are satisfied [8].

3. UPFC Modeling

The Unified Power Flow Controller is a FACTS device designed to control power flow in AC transmission systems. It consists of two voltage source converters: one connected in shunt and the other in series with the transmission line. These converters are interconnected through a common DC link, allowing the exchange of active power.

The shunt converter primarily controls reactive power and maintains the DC link voltage, while the series converter injects a controllable voltage into the transmission line to regulate active power flow as in Figure 1 [9].

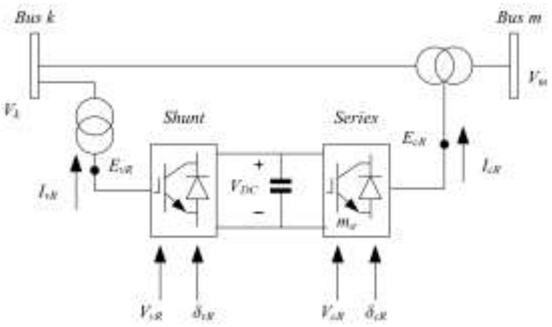

Figure 1. Unified power flow controller (UPFC) system.

For the two voltage sources and constraint equations, suitable formulas would be:

$$E_{vR} = V_{vR} (cos\, \delta_{vR} + J sin\, \delta_{vR}), \tag{1}$$

$$E_{cR} = V_{cR} (cos\, \delta_{cR} + J sin\, \delta_{cR}), \tag{2}$$

$$Re\{E_{vR}\, I_{vR}^{*} + E_{cR}\, I_{m}^{*}\} = 0. \tag{3}$$

From Figure 2 the controllable magnitude and phase angle of the voltage source representing the shunt converter are V (vR) and _vR, respectively.

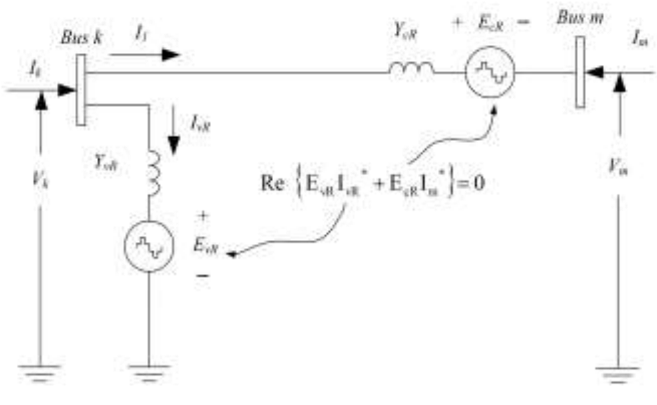

Figure 2. Unified power flow controller equivalent circuit.

$$(V_{vR\,\min} \le V_{vR} \le V_{vR\,\max})$$

$$(0 \le \delta_{vR} \le 2\pi)$$

Between limitations, the amplitude V (cR) and phase angle _cR of the voltage source representing the series converter are adjusted.

$$(V_{cR\,\min} \le V_{cR} \le V_{cR\,\max})$$

$$(0 \le \delta_{cR} \le 2\pi)$$

The operation of the UPFC depends on the magnitude and phase angle of the injected series voltage. By adjusting these parameters, the UPFC can function as a voltage regulator, a phase shifter, and a series compensator [10]. This flexibility enables the UPFC to significantly enhance system performance, improve voltage stability, and reduce transmission losses.

Series converter:

$$P_{cR} = V_{cR}^2 G_{mm} + V_{cR} V_k [G_{km} \cos(\delta_{cR} - \theta_k) + B_{km} sin(\delta_{cR} - \theta_k)] + V_{cR} V_m [G_{mm} \cos(\delta_{cR} - \theta_m) + B_{mm} sin(\delta_{cR} - \theta_m)] \quad (4)$$

$$Q_{cR} = -V_{cR}^2 B_{mm} + V_{cR} V_k [G_{km} \sin(\delta_{cR} - \theta_k) - B_{km} cos(\delta_{cR} - \theta_k)] + V_{cR} V_m [G_{mm} \sin(\delta_{cR} - \theta_m) + B_{mm} cos(\delta_{cR} - \theta_m)] \quad (5)$$

Shunt converter:

$$P_{vR} = -V_{vR}^2 G_{vR} + V_{vR} V_k [G_{vR} \cos(\delta_{vR} - \theta_k) + B_{vR} sin(\delta_{vR} - \theta_k)] \quad (6)$$

$$Q_{vR} = V_{vR}^2 B_{vR} + V_{vR} V_k [G_{vR} \sin(\delta_{vR} - \theta_k) - B_{vR} cos(\delta_{vR} - \theta_k)] \quad (7)$$

The active power supplied to the shunt converter, PvR, equals the active power needed by the series converter, PcR, assuming lossless converter valves; that is,

$$P_{vR} + P_{cR} = 0 \quad (8)$$

As a result, the perceived power exchanged via a series-injected voltage is as follows:

$$S_{cR} = P_{cR} + j\, Q_{cR} = E_{cR} I_m{}^* \quad (9)$$

($P_{cR} > 0$, $Q_{cR} > 0$) means that the system is supplied with both active and reactive power [11].

Shunt current appears to transfer the following electrical energy:

$$S_{vR} = P_{vR} + j\, Q_{vR} = E_{vR} I_{vR}{}^* \quad (10)$$

($P_{vR} > 0$, $Q_{vR} > 0$) means that the system's active and reactive power is being drained. The reactive powers QcR and QvR, on the other hand, can be generated independently by the two converters.

4. Methodology

The novelty of the proposed methodology lies in the coordinated integration of a voltage stability index (Lmn) for candidate location identification with an OPF-based framework for optimal parameter tuning, applied to a real large-scale power system. This combined approach ensures both accurate placement and optimal operation of UPFC devices under realistic system constraints.

4.1. Optimal Placement Using Lmn Index

The Line Stability Index (Lmn) is used to identify weak transmission lines within the network. This index is derived from power flow equations and reflects the proximity of a system to voltage collapse [12].

A value of:

Lmn < 1 → stable system
Lmn ≥ 1 → instability risk

Most line stability indices are based on a single-line power transmission model. In this study, the loading condition is increased proportionally for both active power (P) and reactive power (Q) at all load buses. This approach ensures that the apparent power demand increases consistently while preserving the original power factor characteristics of the system. Figure 3 shows a single line in an interconnected network.

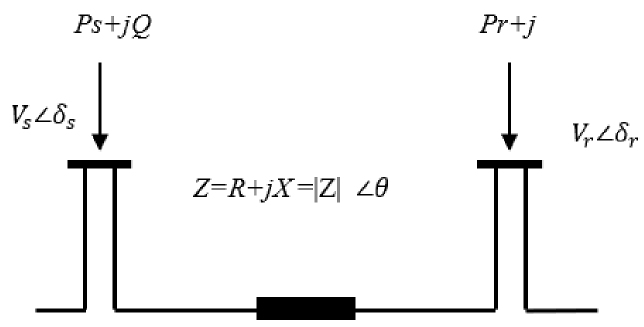

Figure 3. Two bus system.

Where
Vs, Vr: are the sending end and receiving end voltages.
δs, δr: are the phase angles at the sending and receiving end.
Z: is the line impedance.
R: is the line resistance.
X: is the line reactance.
θ: is the line impedance angle.
Ps, Qs: are the active and reactive powers at the sending end.
Pr, Qr: are the active and reactive powers at the receiving end.

This index, proposed in, is based on the concept of power flow over a single line and the reduction of a power system network into a single line [13].

The current I line is determined as follows:

$$I_{line} = \frac{(Vs - Vr)}{Z} \tag{11}$$

The receiving apparent power, given as, can likewise be used to determine the I line.:

$$I_{line} = \left(\frac{S_r}{V_r}\right)^* = \left(\frac{P_r - jQ_r}{V_r\angle -\delta_r}\right) \tag{12}$$

Rearranging Equations (1) and (2) yields:

$$P_r - jQ_r = \frac{|Vs||Vr|}{Z}\angle(\theta - \delta s + \delta r) - \frac{|Vr|^2}{Z}\angle\theta \tag{13}$$

If real and reactive power are separated in this equation, then

$$P_r = \frac{|Vs||Vr|}{Z}cos(\theta - \delta s + \delta r) - \frac{|Vr|^2}{Z}cos\theta \tag{14}$$

$$Q_r = \frac{|Vs||Vr|}{Z}sin(\theta - \delta s + \delta r) - \frac{|Vr|^2}{Z}sin\theta \tag{15}$$

Defining $\delta = \delta s - \delta r$ and solving Equation 3.5 for Vr, then,

$$V_r = \frac{V_s\,sin(\theta-\delta) \pm \sqrt{\left(V_s\,sin(\theta-\delta)\right)^2 - (4XQ_r)}}{2sin(\theta)} \tag{16}$$

If we replace Z sin = X with the constraint that the square root value must be positive, we get,

$$\begin{cases} \left(V_s\,sin(\theta-\delta)\right)^2 - (4XQ_r) \geq 0 \\ L_{mn} = \frac{4XQ_r}{\left(V_s\,sin(\theta-\delta)\right)^2} \leq 1 \end{cases} \tag{17}$$

The system is stable if Lmn is less than one; once Lmn surpasses one, the system approaches its voltage collapse point [14].

The continuation approach has numerous advantages in voltage collapse investigations; thus, most researchers use it to trace voltage profiles at various buses of the test power system with respect to variations in loading level, i.e., continuation power flow [15]. Although voltage stability enhancement and loss minimization are distinct objectives, they are inherently correlated in transmission systems. Lines identified as voltage-weak using the Lmn index typically experience high reactive power stress and inefficient power flow patterns, which also contribute to increased transmission losses. Therefore, improving the operating conditions of these lines through UPFC installation leads to simultaneous enhancement of voltage stability and reduction of system losses.

In this work, the Lmn index is used for candidate selection, while the OPF ensures global optimality in terms of loss minimization, thus resolving the potential conflict between local stability improvement and system-wide efficiency.

To demonstrate the use of the suggested index, each generated and loaded power is steadily increased until one of the LMN values hits an unstable point, at which time the system achieves the stability margin. The value of Lmn fluctuates as the load on the entire load bus changes. At the maximum loading point, the best Lmn values are chosen. MATPOWER is a set of MATLAB 24b M-files that includes a function for power flow continuation. The final selection of UPFC locations is based not only on the ranking of lines according to the Lmn index but also on practical network considerations. Transformer branches were excluded from the candidate set due to their operational characteristics and control limitations. In addition, in radial sections of the network, if multiple consecutive lines exhibited high Lmn values, only the line with the highest index was selected to avoid redundancy. Network topology, operational feasibility, and controllability were therefore considered alongside the Lmn ranking in determining the final set of installation locations. The lines are ranked based on their Lmn values, and those with the highest values are selected as candidate locations for UPFC installation [16]. It should be noted that the proposed approach is based on steady-state optimal power flow analysis and does not represent a closed-loop control system. The UPFC devices are modeled as controllable power injections with predefined setpoints derived from the OPF solution. Therefore, the study focuses on optimal planning and steady-state operation rather than real-time feedback control.

4.2. UPFC Parameter Setting

Optimal Power Flow (OPF) techniques have evolved considerably to meet the increasing operational and planning requirements of modern power systems. The primary objective of OPF is to determine the most suitable operating condition of a power network by optimizing a predefined objective function while satisfying all technical and operational constraints. These constraints generally include the network power balance equations, system loading conditions, voltage magnitude limits, and the generation limits of active and reactive power sources [17].

In the present study, OPF is utilized to identify the optimal steady-state operating condition of the Sudanese national grid. This includes determining the voltage magnitudes and phase angles of all buses, as well as the corresponding optimal active and reactive power flows in the transmission network. The objective function considered in this work is the minimization of total active power losses. The OPF analysis is performed using NEPLAN, which is recognized as a comprehensive and reliable software platform for the simulation, planning, and optimization of transmission, distribution, generation, and industrial electrical networks [18].

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The UPFC control settings are derived from the OPF solution at the selected candidate locations. These settings include the active power flow setpoint, the reactive power flow setpoint, and the voltage magnitude setpoint at the sending-end bus. The feasible operating limits of the UPFC are evaluated using Equations (4)–(10). A proper definition of these limits is necessary to ensure correct device operation; otherwise, the UPFC may fail to regulate the specified active and reactive power flows, which could lead to non-convergence of the power flow solution [19]. The OPF problem in this study is formulated as follows:

Objective Function:

Minimize total active power losses:

$$\min \sum_{\mathrm{k}=1}^{Ni} \mathrm{Ploss,k} \tag{18}$$

Subject to:

Power balance equations (load flow constraints)

- Bus voltage limits:

$$\mathrm{V}_i^{min} \leq V_i \leq \mathrm{V}_i^{max} \tag{19}$$

- Generator limits:

$$P_{Gi}^{min} \leq P_{Gi} \leq P_{Gi}^{max}, Q_{Gi}^{min} \leq Q_{Gi} \leq Q_{Gi}^{max} \tag{20}$$

- Line thermal limits

Decision Variables:

- Generator outputs (P, Q)
- Bus voltages and angles
- UPFC control parameters (series-injected voltage magnitude and angle, shunt reactive current)

Assumptions:

- Steady-state operation
- Lossless converters
- Fixed network topology

This formulation ensures that the UPFC settings are optimally coordinated with system operating conditions.

According to the OPF operating point, the target active and reactive power flows at the candidate lines are established, and the allowable ranges of the injected series voltage VcR and shunt reactive current Iq are adjusted to achieve the desired power flow profile. Based on these considerations, the following assumptions are made for all installed UPFC devices:

(a) The source impedances of the UPFC devices are neglected.
(b) The magnitude of the injected series voltage VcR is assumed to vary within 0–0.3 p.u.
(c) The shunt reactive current Iq is assumed to vary within 0–190 A.
(d) The maximum power exchange through the DC link is limited to 50 MW.

All branches with transformers were not evaluated in the UPFC installation procedure for optimal placement. The optimum position for a UPFC is a line with the highest Lmn value, followed by other lines with lower values. If two or more cascaded lines in a radial network are regarded as the best location, the one with the highest Lmn value is chosen as the optimal UPFC placement position. The UPFCs are inserted one at a time, and the system improvements are compared to the prior example until no improvements are found. The proposed approach is depicted in Figure 4 as a flowchart [20]. The relationship between the Lmn index and voltage stability margin is based on its ability to indicate proximity to voltage collapse. As the system loading increases, the Lmn value approaches unity, indicating a reduction in stability margin. This behavior has been validated in previous studies and is widely used as an indicator of voltage instability.

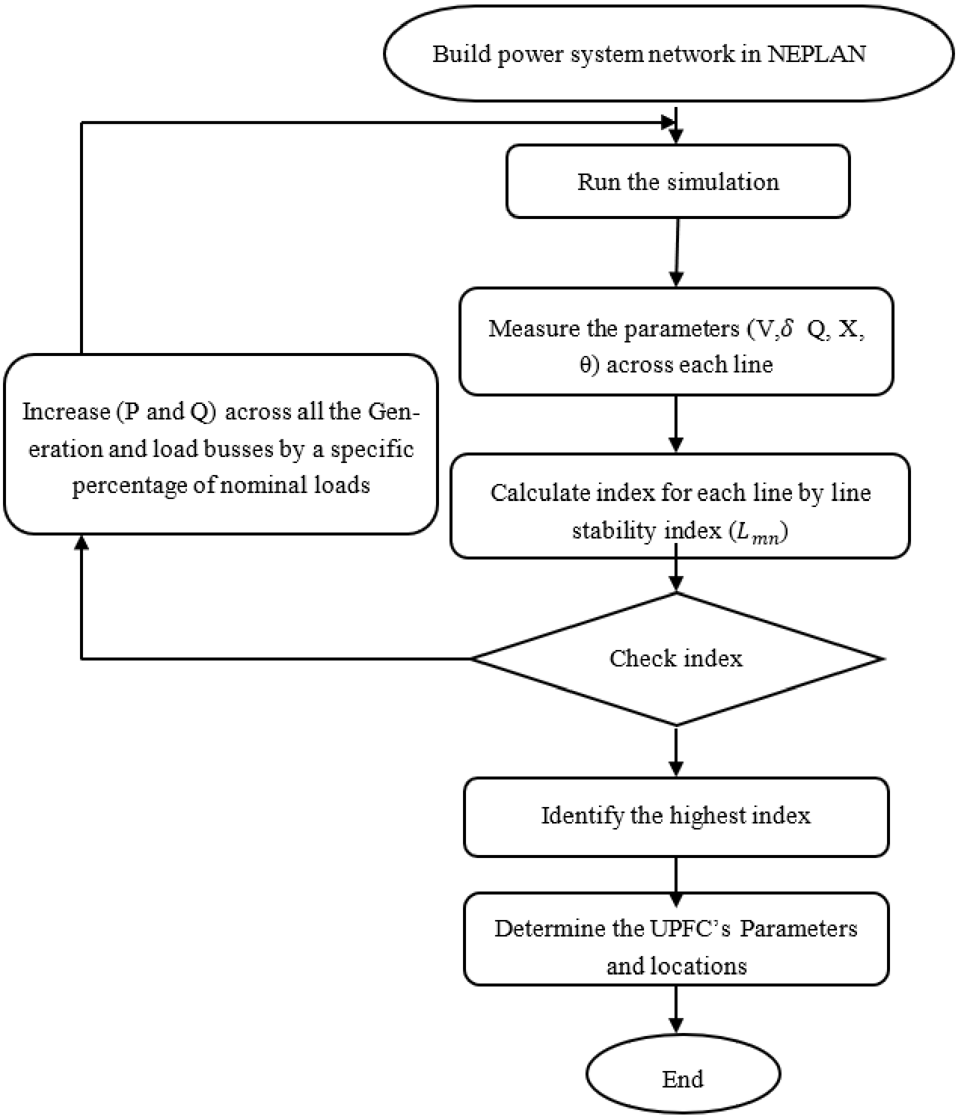

Figure 4. Flowchart of the proposed methodology highlighting the integrated Lmn–OPF framework for optimal UPFC placement and parameter tuning.

5. Case Study and Results

As a test power network, the real Sudanese electrical power grid was chosen. In terms of design and actual difficulties, this regional network can be assumed to be representative of the entire country's network. The reported voltage profile improvement of 35.29% is calculated based on the reduction in the total voltage deviation index, defined as

$$VD = \sum_{i=1}^{Nb} |V_i - 1.0| \tag{21}$$

where Vi represents the per-unit voltage magnitude at bus iii. The percentage improvement is computed as

$$\text{Improvement} = \frac{VD_{base} - VD_{UPFC}}{VD_{base}} \times 100\% \tag{22}$$

Figure 5 shows a simplified single-line schematic of the network.

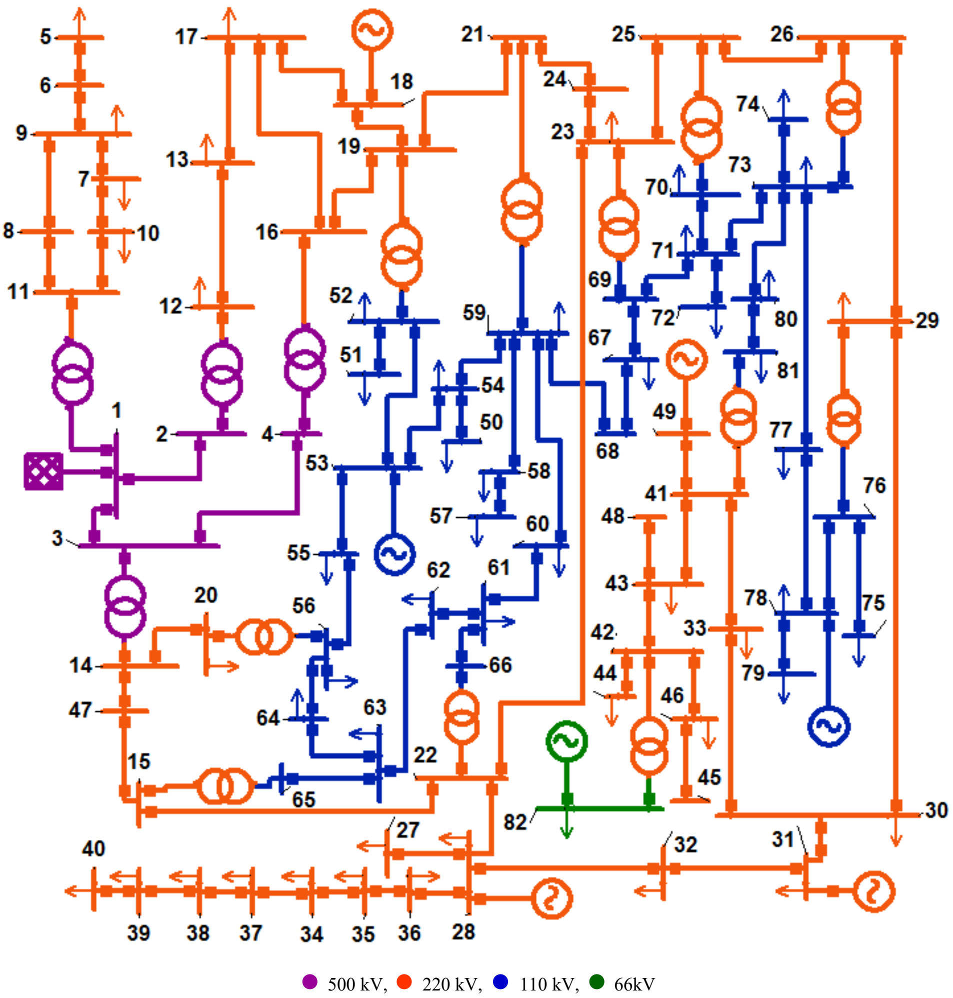

Figure 5. Single-line diagram of Sudanese electrical network.

Table 1 depicts network characteristics for data collected under normal load conditions.

Table 1. Network characteristics.

Buses	82	Include 500, 220, 110 and 66 kV level.	
Lines	81	Totaling 5469.735 km long.	
Power plants	7	Merowe (slack) Garri Roseires Khartoum North Rabak Sennar Girba	total power generated is 1603.517 MW, 227.008 MVAr
Tie line	1	With Ethiopian power network.	
Loads	53	total loading level is 1553.3 MW, 1083.3 MVAr	
Shunts	10	total +15 MVAr inductive −630 MVAr capacitive	
Transformers	15	Total rating is 5810 MVA	

Optimal Locations and Parameter Setting

The system's generation and load are steadily increased until the collapse point is reached, allowing the unstable lines to be verified under high loading conditions. (max) = 1.94 is the maximum loading point [21].

The lines are sorted from highest to lowest L $_{MN}$ value, with the most severe lines in the system being given in Table 2. The candidate locations for UPFC installation are represented in this diagram [22].

Table 2. Selected places for UPFC installation.

Rank	Line	From Bus	To Bus	L_{mn} at λmax = 1.94
1	7	MWT2	DEB2-B1	1.000164244
2	9	DEB2-B2	DON2	0.90185929
3	5	ATB2	SHN2	0.645191638
4	40	OHAS1	GAD B2	0.623671911
5	25	KLX1	KUK1	0.549721636
6	60	UMR2	OBD2	0.413167239
7	53	SNP1	MIN1	0.382504059

The values of the optimal power flow are used to set the power flow parameters for the UPFCs. Table 3 displays the control values for the active and reactive line flow, as well as the control value for the voltage magnitude of the transmitting bus for the seven UPFC devices [23].

Table 3. Parameters for installed UPFC devices.

Name	Line	P Set	Q Set	Vs Set
		MW	MVAr	V
UPFC-1	7	−53.4	−11.1	1
UPFC-2	9	−43.7	−17.3	1.026
UPFC-3	5	−0.4	−121.5	0.954
UPFC-4	40	9.6	31.5	1.007
UPFC-5	25	142.1	132.7	0.986
UPFC-6	60	38.2	−15	1
UPFC-7	53	−14	8.2	0.998

System improvements after installing the UPFC devices:

Table 4 shows the bus voltages and angles after the UPFCs were installed, and Table 5 shows the load flow result after the UPFCs were installed [24].

Table 4. Bus voltage and angle after installing UPFC devices.

Bus Number	Bus Name	Per-Unit Voltage		Angle Degree	
		With	Without	With	Without
1	MWP500	1	1	0	0
2	ATB5	1.0213	1.0031	−1.6	−3.7
3	MRK5	0.9691	0.9777	−9.2	−7.6
4	KAB5	0.9708	0.9799	−10.1	−8.3
5	WHL2	0.9895	0.9892	−5.3	−8.7
6	WWA2	1.0208	1.0204	−4.6	−8
7 *	DEB2-B1	1	1.0216	−2.6	−6.2
8	DEB2-B2	1.038	1.0443	−5.1	−5
9 *	DON2	1.026	1.0256	−4.3	−7.7

10	MWT2	1.0181	1.0315	−2.7	−2.8
11	MWP2	1.0327	1.0384	−1.5	−1.5
12 *	ATB2	0.954	0.924	−3.7	−9.1
13	SHN2	0.9696	0.9688	−21.8	−13.8
14	MRK2	1.0007	1.0051	−22.8	−19.4
15	GAM2	0.9807	0.9844	−23.9	−20.4
16	KAB2	0.9851	0.9871	−21.1	−16.6
17	FRZ2	0.9974	0.9976	−21.5	−16.4
18	GER2	1	1	−21.5	−16.4
19	IBA2	0.9922	0.9951	−23.1	−18.6
20	MHD2	1.0002	1.0045	−23.5	−20
21	KLX2	0.9901	0.9938	−23.7	−19.4
22	JAS2	0.9771	0.9812	−24.5	−20.8
23	GAD2	0.9759	0.9831	−24.9	−21
24	SOB2	0.987	0.9914	−24	−19.8
25	NHAS2	0.9689	0.9574	−27.2	−23.5
26	MAR2	0.9637	0.9559	−27.8	−24.2
27	MSH2	0.9943	0.996	−24.3	−20.6
28	RBK2	1	1	−23.8	−20.1
29	SNJ2	0.9755	0.971	−27.3	−23.7
30	SNG2	0.9923	0.9893	−26.7	−23.1
31	ROS2	1	1	−22.1	−18.5
32	RNK2	1.0079	1.0079	−23	−19.4
33	HWT2	1.0005	0.9983	−27.3	−23.7
34	OBD2	0.9887	0.9987	−25.1	−23.2
35 *	UMR2	1	1.0054	−25.2	−21.6
36	TND2	1.004	1.0072	−24.7	−21.1
37	DBT2	0.9914	1.0019	−25.8	−23.9
38	ZBD2	0.9899	1.0006	−26.1	−24.1
39	FUL2	0.9917	1.0026	−26.5	−24.6
40	BBN2	0.9937	1.0046	−26.6	−24.7
41	GDF2	0.9996	0.9983	−27.7	−24.2
42	GRB2	0.9927	0.9916	−28.2	−24.6
43	SHK2	0.9992	0.9978	−27.8	−24.2
44	NHLF2	0.9925	0.9914	−28.3	−24.7
45	ARO2	0.9994	0.9982	−28.5	−25
46	KSL2	0.9984	0.9972	−28.5	−25
47	HUD	0.9959	1.0002	−23	−19.6
48	UTP2	0.9996	0.9983	−27.8	−24.2
49	SHD2	1	1	−26.5	−23
50	KHE1	0.9833	0.9821	−33.5	−26.8
51	IZB1	0.9661	0.9663	−31.6	−25.7
52	IBA1	0.9733	0.9734	−31.1	−25.3
53	KHN1	1	1	−32.5	−26.4
54 *	KUK1	0.986	0.9848	−33.3	−26.6
55	IZG1	0.9777	0.975	−31.6	−26.3
56	MHD1	0.969	0.9642	−30.6	−25.9
57	FAR1	0.9547	0.9288	−29.2	−26.8
58	AFR1	0.9621	0.9364	−28.8	−26.4
59	KLX1	0.9694	0.944	−28.4	−26
60	LOM1	0.9674	0.9441	−28.8	−26.2
61	SHG1	0.9695	0.9516	−29.3	−26.3
62	MUG`	0.967	0.9547	−30.1	−26.5

63	BNT1	0.9688	0.9585	−30.2	−26.4
64	OMD1	0.9658	0.9576	−30.6	−26.4
65	GAM1	0.9937	0.9858	−29.2	−25.5
66	JAS1	1.0133	1.0035	−27.9	−24.7
67	BAG1	0.9983	0.9203	−28.9	−24
68	SOB1 B2	0.9798	0.9355	−28.6	−25.3
69 *	GAD B2	1.007	0.924	−28.8	−23.6
70	NHAS1	1.0275	1.0349	−28.7	−25.7
71	OHAS1	1.0277	1.0317	−28.9	−25.9
72	GND1	1.0244	1.0284	−29.1	−26.1
73	MAR1 B1	1.0334	1.0291	−30.3	−26.8
74	MAN1	1.023	1.0186	−31.2	−27.6
75	ORBK1	1.0205	1.0197	−28.7	−25.2
76	SNJ 1	1.0209	1.0201	−28.6	−25.1
77	HAG1	1.0143	1.0116	−29.8	−26.3
78	SNP1	1	1	−28.2	−24.8
79 *	MIN1	0.998	0.9481	−28.2	−25.8
80	FAO1	1.0085	1.0049	−30.7	−27.2
81	GDF 1	1.0182	1.0165	−29.1	−25.5
82	GRB6	1	1	−29	−25.5

* denotes the controlled voltage bus by UPFCs; ▒ denotes the buses near to the voltage limits.

Table 5. Power flow results after installing UPFC devices.

Line	From Bus	To Bus	P MW	Q MVAr	P Loss MW		Q Loss MVAr	
					With	Without	With	Without
1	69	67	46.6	45.7	0.4	0.2	0.3	0.2
2	6	5	14.5	0.6	0.2	0.2	−40.3	−40.3
3	7	9	21.4	−35.6	0.2	0.1	−18.3	−19.5
4	11	10	85.2	34.7	0.4	0.4	−2.7	−3.1
5 *	12	13	1.5	76.6	1.1	4.0	−44.9	−32.0
6	11	8	44.7	−23.4	0.5	0.5	−22.7	−23.0
7 *	10	7	1.6	5.7	0.1	0.6	−19.2	−16.8
8	9	6	14.6	−26.7	0.0	0.0	−68.9	−68.9
9 *	8	9	44.1	−0.7	0.4	0.4	−18.0	−18.3
10	1	2	101.4	−211.9	0.5	1.6	−243.6	−227.9
11	1	3	826.7	−212.0	13.6	9.3	−554.9	−604.1
12	3	4	362.3	−91.8	0.6	0.4	−29.7	−32.7
13	13	17	35.3	55.3	0.6	2.0	−41.8	−36.2
14	17	18	18.8	178.8	0.1	0.1	−1.5	−1.5
15	16	19	331.0	−147.5	2.8	2.9	0.9	1.2
16	52	51	68.2	39.8	0.2	0.2	−0.2	−0.2
17	52	53	123.3	−223.9	2.3	1.9	8.0	6.3
18	54	50	82.0	53.2	0.1	0.1	0.0	0.0
19	56	55	149.0	−131.6	0.9	0.4	3.0	0.9
20	53	54	269.6	216.5	0.9	0.6	6.4	4.2
21	53	55	73.8	−181.2	1.3	1.1	4.1	3.3
22	14	20	219.1	−48.5	0.7	0.6	−5.4	−6.2
23	58	57	50.4	32.6	0.2	0.2	−0.7	−0.6
24	59	58	67.8	40.9	0.2	0.2	−0.2	−0.1
25 *	59	54	142.1	132.7	2.0	1.7	7.5	6.5
26	59	60	181.4	12.5	0.3	0.1	0.9	0.1

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27	60	61	95.7	−46.5	0.3	0.2	0.3	0.0
28	61	62	105.0	−6.1	0.4	0.0	0.4	−0.8
29	63	62	29.2	−49.1	0.0	0.1	−0.2	0.0
30	56	64	4.1	−31.1	0.0	0.2	−0.8	0.1
31	64	63	85.6	23.3	0.1	0.0	0.0	−0.5
32	65	63	116.6	105.8	1.2	1.3	3.1	3.6
33	21	24	120.5	51.0	0.1	0.2	−1.9	−1.7
34	24	23	120.4	53.0	0.5	0.7	−6.2	−5.4
35	67	68	9.7	21.1	0.3	0.5	−0.2	0.1
36	68	59	9.4	21.4	0.2	0.3	−0.1	0.1
37	66	61	75.8	82.4	1.4	1.9	1.6	3.7
38	22	23	57.1	−9.1	0.1	0.0	−13.3	−13.7
39	23	25	106.7	−10.0	0.8	1.3	−18.0	−15.3
40 *	71	69	9.6	20.3	1.0	2.4	−1.6	0.5
41	70	71	37.4	−13.3	0.0	0.1	−0.4	−0.3
42	71	72	27.9	12.8	0.0	0.0	−1.4	−1.4
43	71	73	7.0	−9.7	0.2	0.1	−1.7	−1.8
44	27	22	20.2	7.6	0.2	0.1	−56.3	−56.8
45	28	27	27.2	−8.9	0.1	0.1	−42.1	−42.2
46	25	26	43.7	6.9	0.1	0.1	−13.6	−13.3
47	73	74	11.0	1.7	0.1	0.1	−2.3	−2.3
48	29	26	27.3	16.8	0.1	0.1	−21.1	−20.6
49	73	77	3.2	13.0	0.2	0.2	−1.0	−1.0
50	78	77	4.8	−11.6	0.2	0.2	−1.7	−1.7
51	76	78	17.6	46.7	0.7	0.6	0.5	0.4
52	76	75	0.3	−1.8	0.0	0.0	−3.3	−3.3
53 *	78	79	0.4	−0.6	0.0	0.6	−2.3	−1.4
54	73	80	6.9	3.5	0.1	0.1	−2.3	−2.2
55	30	29	62.1	62.3	0.3	0.4	−11.5	−11.1
56	31	30	105.5	−29.7	1.6	1.6	−40.2	−39.8
57	31	32	34.2	−56.3	0.2	0.2	−55.3	−55.3
58	30	33	29.8	−57.4	0.2	0.2	−34.9	−34.6
59	28	36	44.5	−45.7	0.2	0.2	−43.5	−43.4
60 *	35	34	1.4	3.2	0.0	0.3	−24.6	−24.0
61	34	37	19.8	−18.9	0.1	0.1	−17.1	−17.4
62	37	38	13.5	−5.8	0.0	0.0	−12.8	−13.1
63	38	39	10.1	−16.7	0.0	0.0	−23.1	−23.6
64	39	40	3.4	−12.9	0.0	0.0	−15.4	−15.8
65	36	35	43.3	−7.8	0.1	0.1	−30.8	−31.1
66	32	28	33.6	−23.4	0.1	0.1	−64.7	−64.8
67	33	41	28.1	−23.5	0.1	0.1	−39.5	−39.3
68	42	46	16.7	−43.8	0.1	0.1	−37.2	−37.1
69	43	42	42.1	8.8	0.1	0.1	−27.1	−27.1
70	41	43	42.8	7.0	0.0	0.0	−3.2	−3.2
71	46	45	0.0	−17.3	0.0	0.0	−17.3	−17.3
72	42	44	12.0	−11.1	0.0	0.0	−19.1	−19.1
73	14	47	231.7	130.9	0.4	0.4	−1.8	−2.0
74	47	15	231.3	132.6	1.4	1.3	−5.2	−5.9
75	17	16	30.6	−129.4	0.4	0.3	−13.1	−13.8
76	43	48	0.0	−11.1	0.0	0.0	−11.1	−11.1
77	18	19	167.2	−4.6	1.2	2.0	−19.0	−15.6
78	19	21	261.4	−12.7	0.7	1.3	−2.8	−0.5
79	49	41	38.0	−46.9	0.2	0.2	−76.2	−76.1

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80	81	80	4.4	−4.5	0.1	0.1	−5.2	−5.2
81	15	22	113.2	10.4	0.3	0.2	−9.5	−10.0
Total					**46.6929**	**50.1895**	**−1928.76**	**−1943.2**

* denotes UPFC locations.

The power flows in the UPFC-upgraded network differ from the initial situation, as expected. Figure 6 presents the most notable variations in voltage profile [25].

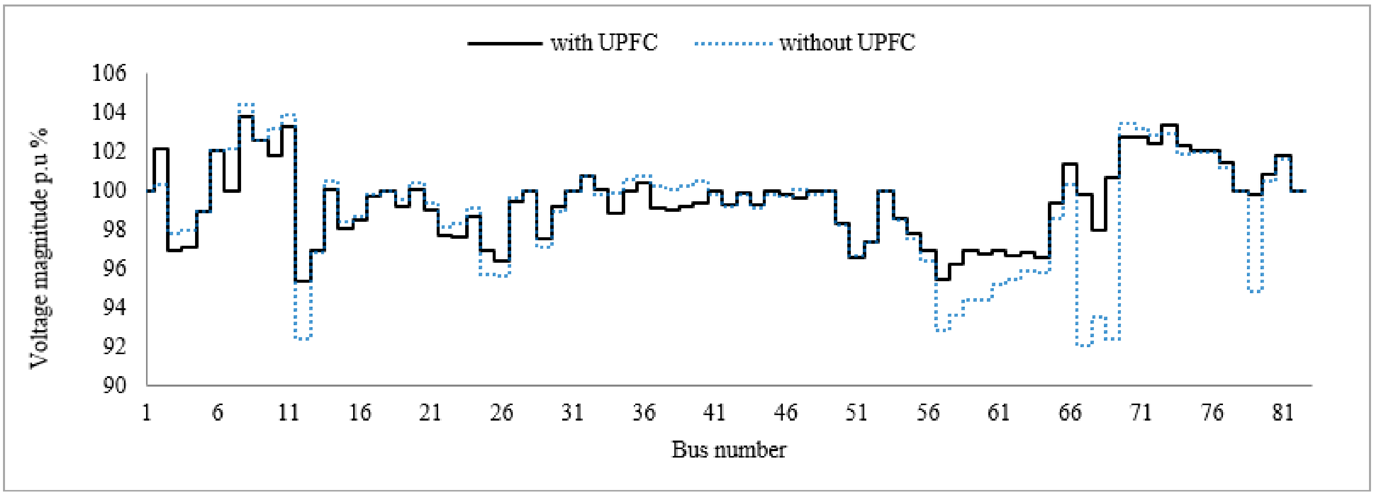

Figure 6. Voltage profiles with and without UPFC.

The Sudanese national electrical utility defines the acceptable voltage operating limits within ±10% of the nominal voltage. This regulatory margin ensures safe and reliable operation of electrical equipment while maintaining system stability under varying loading conditions.

The simulation results reveal a substantial improvement in the overall voltage profile of the network after the installation of UPFC devices, with an enhancement reaching approximately 35.29% compared to the base case. Although all bus voltages in the base case remain within the permissible statutory limits, it is important to note that six buses were operating close to the lower voltage boundary, indicating a high risk of voltage instability under increased loading conditions [26].

As illustrated in Figure 7, the installation of UPFC devices leads to a noticeable improvement in the voltage levels of all weak buses. This improvement can be attributed to the UPFC's capability to provide dynamic reactive power compensation and regulate voltage magnitude through its shunt and series converters.

More specifically, the maximum bus voltage in the system is observed at bus 8 (DEB2-B2), which slightly decreases from its base value by 0.63%, resulting in a final value of 1.038 p.u. This slight reduction is beneficial, as it brings the voltage closer to the nominal value and prevents potential overvoltage conditions, thereby improving system reliability [27]. The observed change in voltage angle at bus SHN2 is attributed to the power flow redistribution caused by the UPFC devices. Since the UPFC directly controls both active and reactive power flow, it can significantly influence phase angles in the network. The change is consistent with the modification of power transfer paths and does not violate system stability constraints, as confirmed by successful power flow convergence and maintained voltage limits.

On the other hand, the minimum voltage level, which represents the weakest point in the network, shows a significant improvement. At bus 12 (ATB2), the voltage increases by approximately 3%, rising from a critically low value of 0.924 p.u. in the base case to

0.954 p.u. after UPFC installation. This improvement is particularly important, as low-voltage buses are typically associated with voltage instability and may lead to voltage collapse if not properly mitigated.

From a system-wide perspective, all bus voltages are maintained within a tighter range of ±5% of the nominal voltage, which is considerably better than the allowable ±10% limit. This indicates a more stable and well-regulated network.

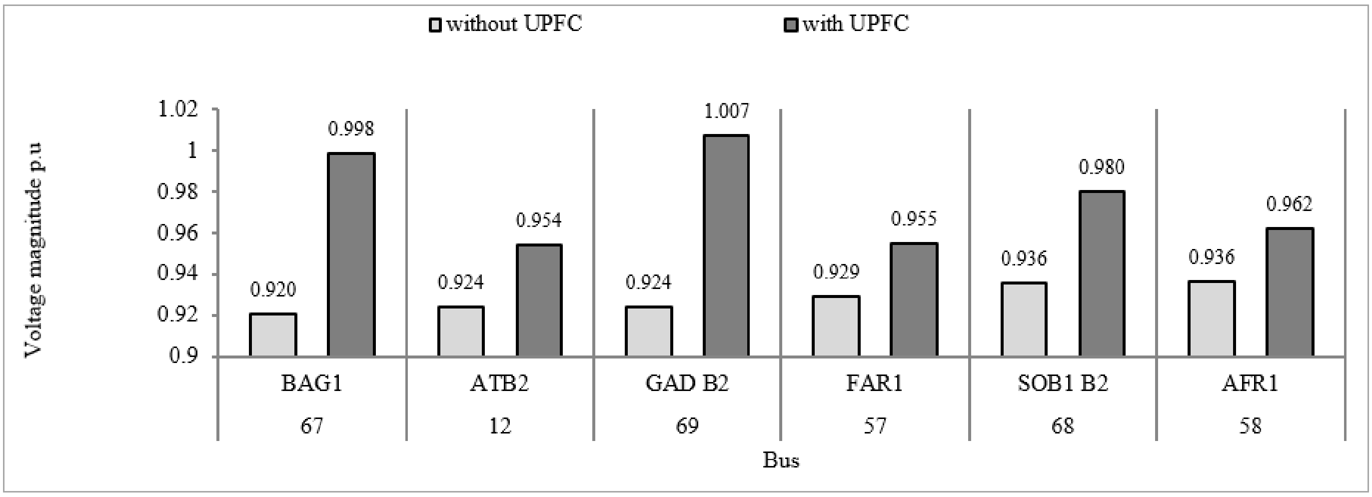

Figure 7. Voltage improvements at weak buses.

Figure 8 presents the percentage loading of transmission lines under normal operating conditions, both before and after the installation of UPFC devices. In the base case, six transmission lines are loaded above 50% of their maximum thermal capacity, indicating the presence of moderate congestion within the network.

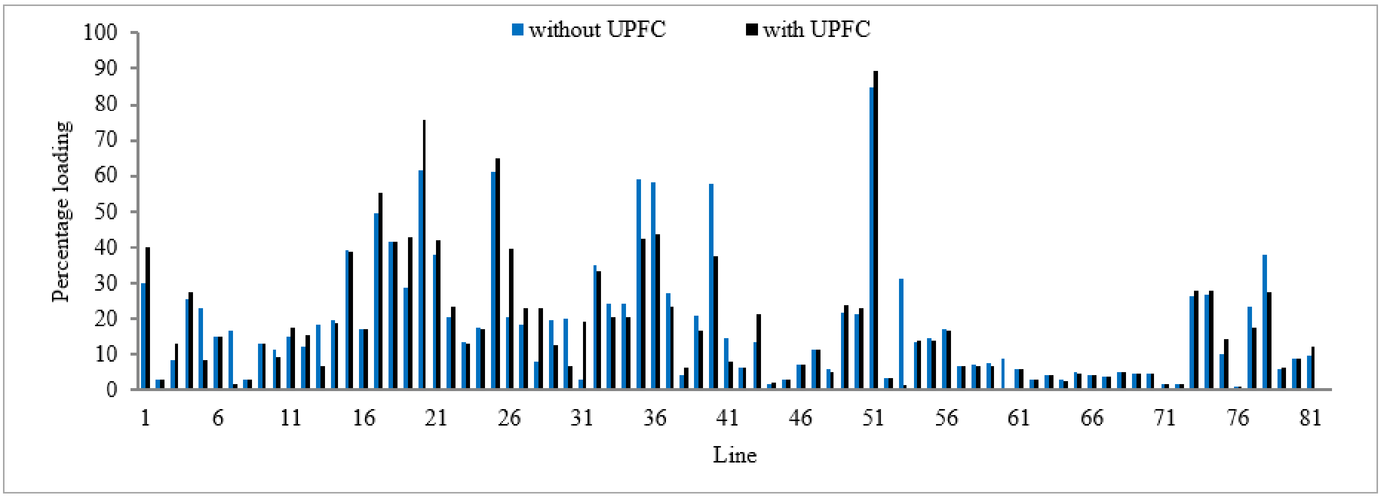

Figure 8. Percentage loading with and without UPFC.

Following the installation of UPFC devices, a noticeable redistribution of power flow occurs. The number of lines loaded above 50% is reduced to four, reflecting an improvement in network loading conditions. In particular, the loading levels of 45 transmission lines decrease, demonstrating that the UPFC devices effectively balance power flow across the system [28].

Although line 11 (MWP500–MRK5) continues to carry the highest power flow in the network, its loading is significantly reduced to 17.63% of its maximum rating. This indicates that the UPFC successfully alleviates stress on heavily loaded lines by redirecting power to alternative paths.

On the other hand, line 51 (SNJ1–SNP1) exhibits the highest loading level of 89.38%, with a slight increase of 4.4% compared to the base case. This increase is attributed to the redistribution of power flow caused by the UPFC devices. Despite this rise, the loading remains within acceptable limits and does not adversely affect system performance [29]. A sensitivity analysis was conducted by incrementally increasing the number of installed UPFC devices (from 1 to 8). The results indicate that performance improvement follows a diminishing return pattern. While the transition from 1 to 7 devices yields significant improvements, the addition of an eighth device results in negligible gains (<1%). Therefore, seven UPFCs represent a near-optimal solution based on performance saturation.

Overall, the total system loading is reduced by 3.26%, confirming that UPFC installation leads to a more efficient and balanced utilization of the transmission network.

Figures 9 and 10 illustrate the distribution of active and reactive power losses across the network before and after the installation of UPFC devices.

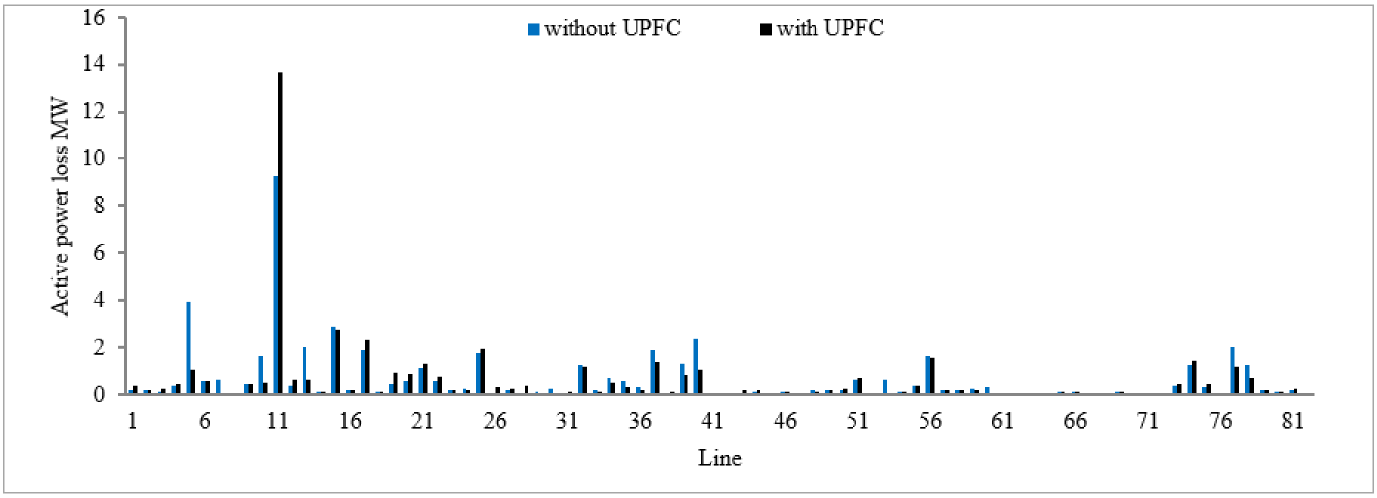

Figure 9. Active power losses with and without UPFC.

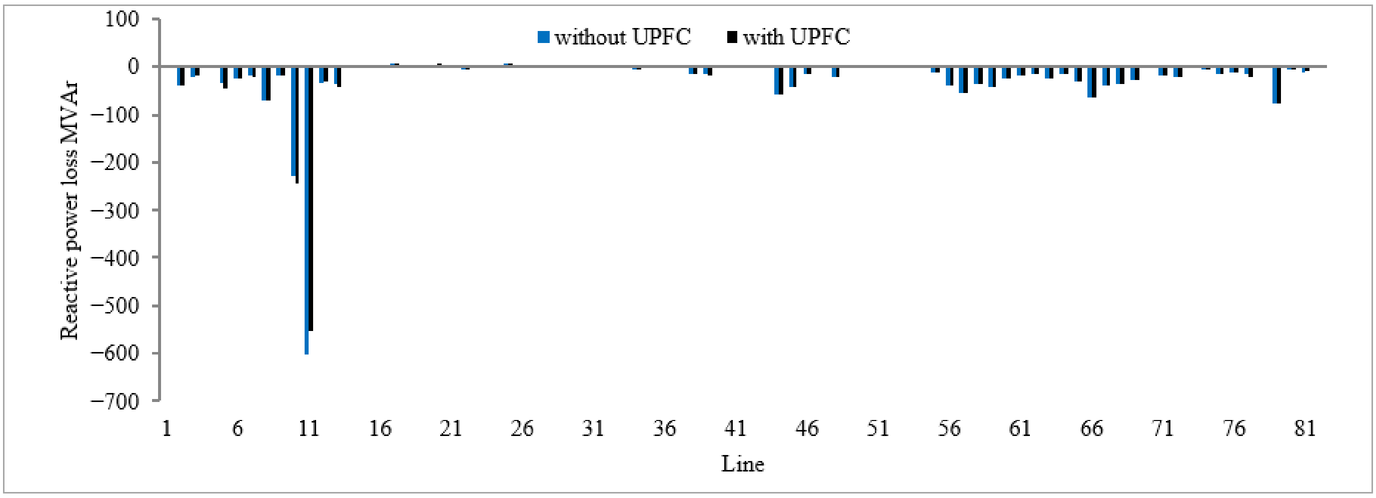

Figure 10. Reactive power losses with and without UPFC.

In the base case, the highest losses occur in lines connected to the slack bus, where the power injection into the system is greatest. Line 11 (MWP500–MRK5) experiences the largest losses, with active and reactive power losses of 9.2555 MW and −604.129 MVAr, respectively. This is primarily due to the high-power flow through this line [30].

After the integration of UPFC devices, the total active power losses of the system decrease to 46.6925 MW, corresponding to an overall reduction of approximately 3.4966 MW. The most significant improvement is observed at line 5 (ATB2–SHN2), where UPFC-3 is installed, resulting in a reduction of 2.8714 MW in active power losses.

Although the active power loss in line 11 increases by 4.3868 MW, this localized increase does not negatively impact the overall system performance. Instead, it reflects the redistribution of power flow, where losses are shifted between lines while the total system losses are reduced.

Similarly, the total reactive power losses decrease to 1928.76 MVAr, indicating improved reactive power management. Line 11 still exhibits the highest reactive losses (554.905 MVAr), but it also shows the largest reduction of 49.224 MVAr, highlighting the effectiveness of UPFC devices in controlling reactive power flow and improving voltage support.

A comparative analysis was conducted for different numbers of installed UPFC devices (1, 3, 5, 7, and 8). The results indicate that system performance improves progressively with the number of devices; however, the rate of improvement decreases significantly beyond seven devices. While the addition of an eighth UPFC results in only marginal improvements in voltage profile and loss reduction, it introduces additional cost and system complexity. Therefore, seven UPFC devices are considered the optimal compromise between performance enhancement and economic feasibility. Figure 11 shows the variation in voltage profiles as a function of the number of installed UPFC devices. The results indicate that voltage levels improve progressively as more devices are introduced into the system. Weak buses benefit significantly from this improvement, moving closer to their nominal voltage values.

However, the improvement becomes less pronounced beyond a certain number of devices. For example, when an eighth UPFC device is installed, the voltage profile shows only a slight improvement compared to the case with seven devices. This suggests that the system has reached a near-optimal operating condition.

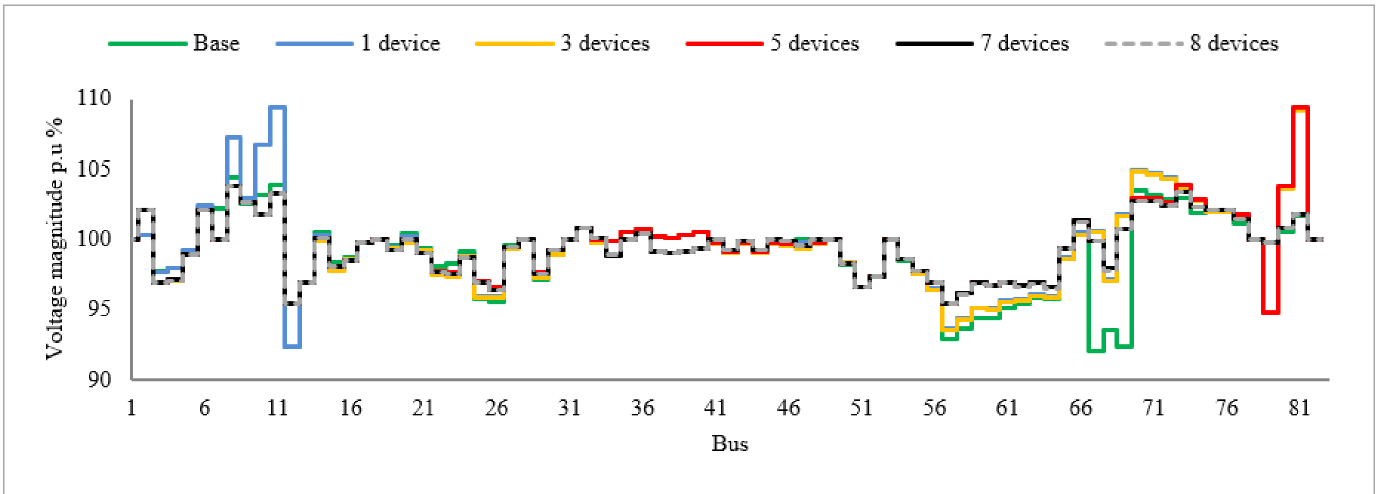

Figure 11. Voltage profiles according to the number of UPFC devices installed.

Figures 12 and 13 further illustrate the relationship between the number of UPFC devices and system performance. Both active and reactive power losses decrease as additional UPFC devices are installed. Nevertheless, the rate of improvement diminishes with each additional device, indicating a diminishing return effect.

These results confirm that there exists an optimal number of UPFC devices, seven in this study, beyond which further installations provide only marginal benefits.

Figure 14 presents the total active power losses in the system before and after installing the UPFC devices. In the base case, the total active power loss is approximately 50.2 MW [31]. After the installation of the UPFC devices, this value is reduced to about 46.7 MW. This reduction of nearly 3.5 MW indicates a clear improvement in system efficiency, which can be attributed to better control of power flow and reduced current circulation in transmission lines.

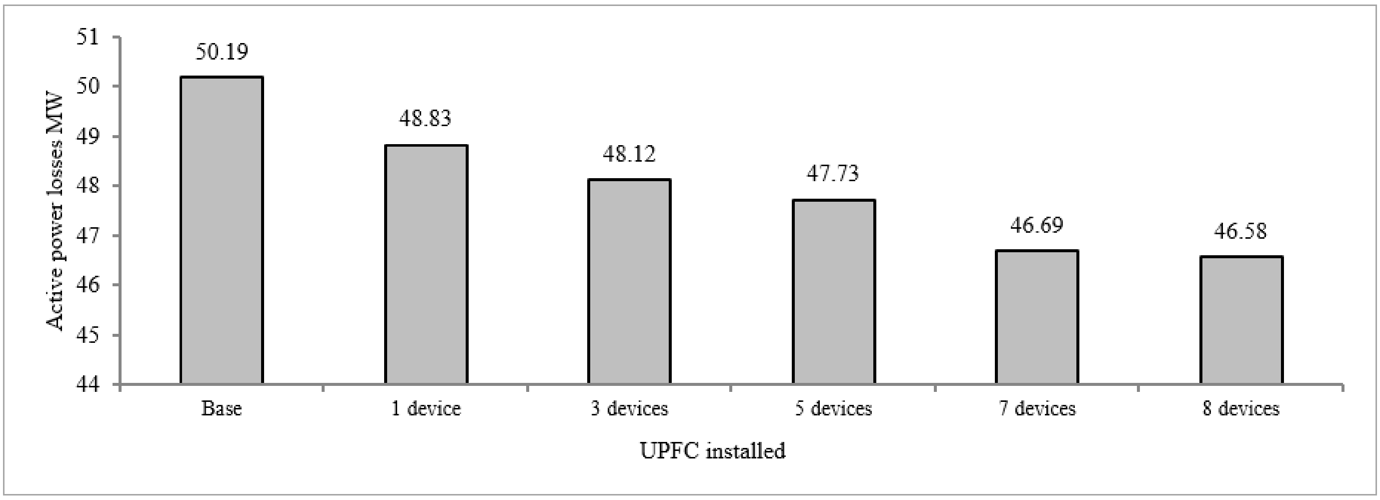

Figure 12. Active power losses according to number of devices installed.

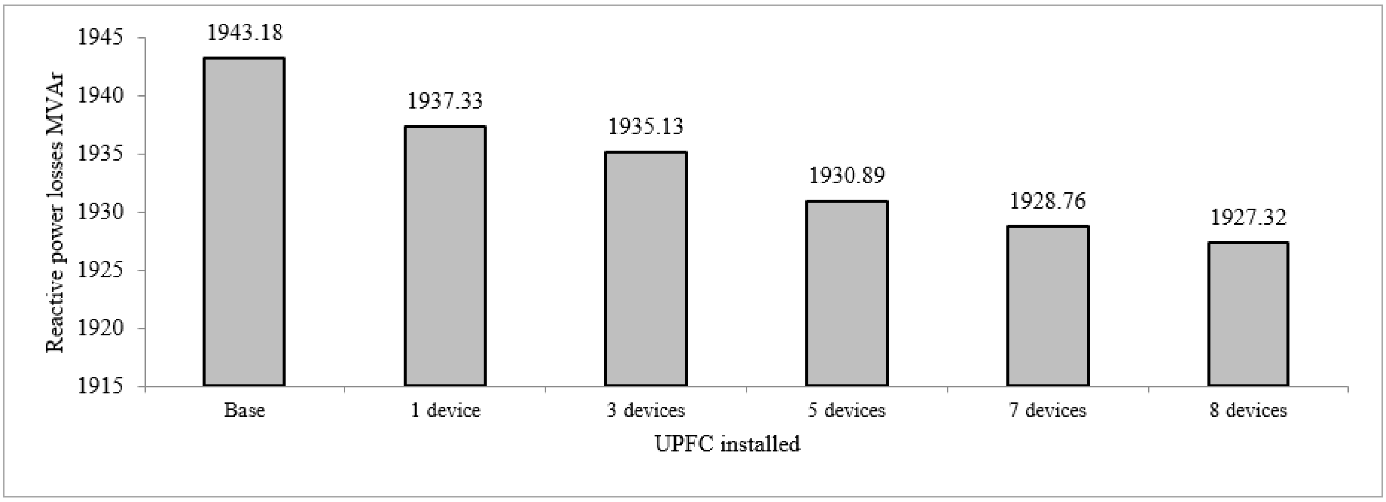

Figure 13. Reactive power losses according to number of devices installed.

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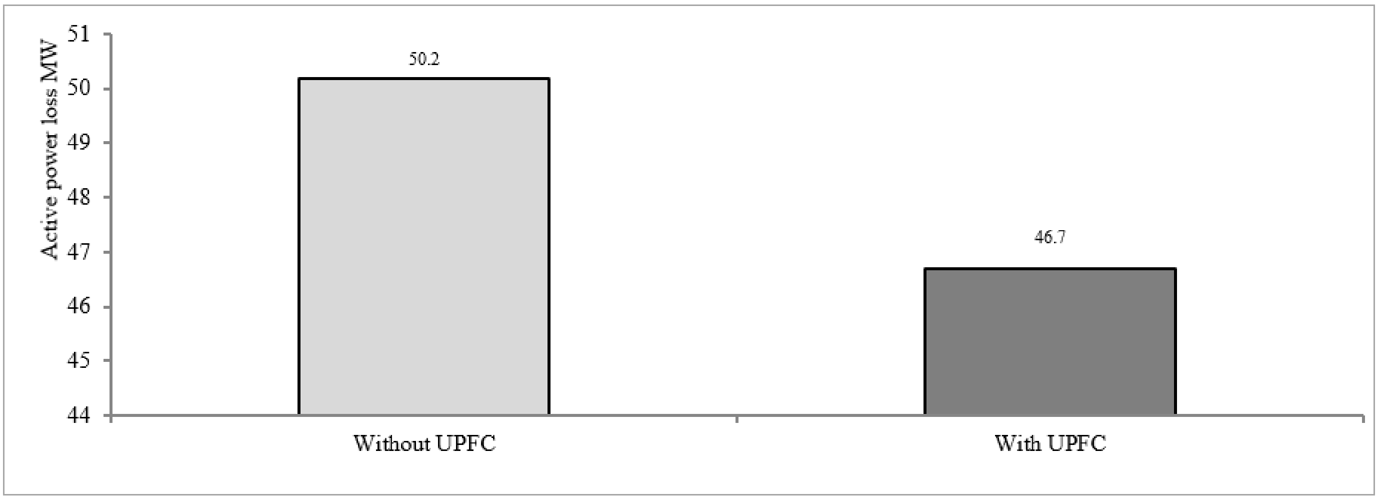

Figure 14. Total active power loss.

Figure 15 shows the total reactive power losses in the system. The results indicate that reactive power losses decrease from approximately −1943 MVAr in the base case to −1928.8 MVAr after UPFC installation. This reduction of about 14.4 MVAr reflects improved reactive power management, as the UPFC provides local compensation and reduces the need for reactive power transfer over long distances.

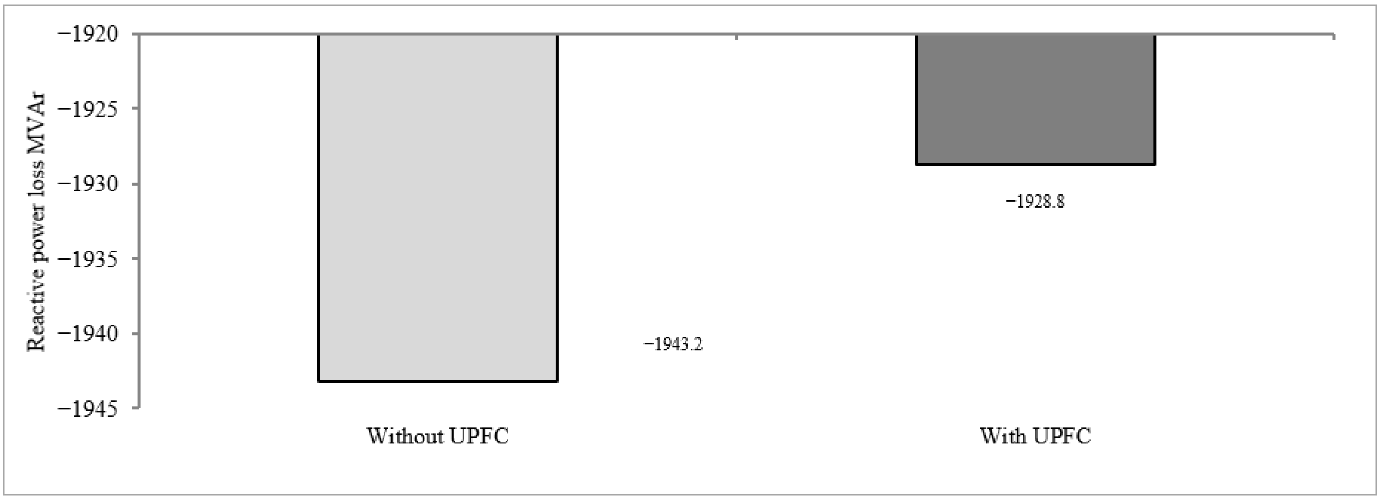

Figure 15. Total reactive power loss.

Figure 16 illustrates the total saved power and the percentage improvement achieved in the system. The results show that the saved active power is about 3.50 MW, while the saved reactive power is approximately 14.42 MVAr. In terms of percentage, the reduction in active power losses is about 6.97%, whereas the reduction in reactive power losses is around 0.74%.

Overall, these results confirm that the installation of UPFC devices leads to a significant reduction in system losses and enhances the overall efficiency of the power network [32]. The improvement in active power losses is more pronounced compared to reactive power losses, indicating that the UPFC is particularly effective in optimizing real power flow while still contributing to better voltage support through reactive power compensation.

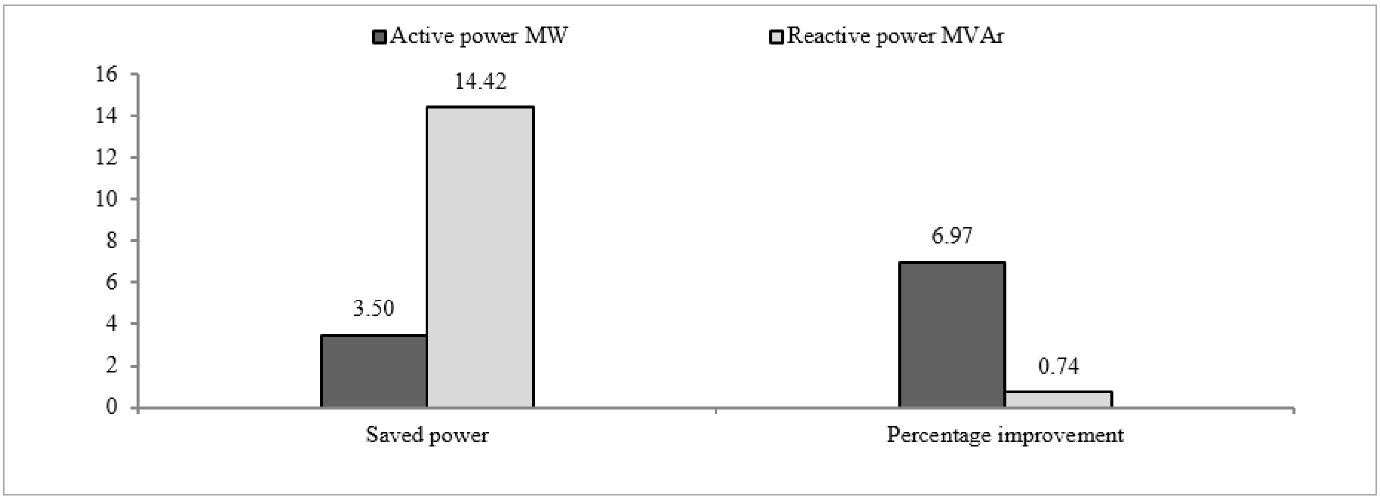

Figure 16. Total saved power and percentage improvement.

6. Conclusions

This study presents an effective methodology for the optimal placement of Unified Power Flow Controller (UPFC) devices with the primary objective of minimizing transmission line losses and preventing bus voltage limit violations. The proposed approach was applied to a real-world case study based on the Sudanese national electrical grid, which represents a large-scale and practically constrained power system.

The Line Stability Index (Lmn), evaluated under heavy active and reactive loading conditions, was employed to identify the most critical transmission lines and determine the optimal locations for UPFC installation. In addition, an Optimal Power Flow (OPF) formulation was developed to calculate the appropriate control parameters of the UPFC devices, with the objective of minimizing total active power losses while satisfying system constraints. The simulation and analysis were carried out using NEPLAN software.

The results demonstrate that the optimal number of UPFC devices for the studied system is seven. The installation of these devices leads to significant improvements in overall system performance. Specifically, the voltage profile is enhanced by 35.29%, and all bus voltages are maintained within a tighter range of ±5% of the nominal value, thereby improving voltage stability and operational reliability.

Furthermore, the integration of UPFC devices results in measurable energy savings, with reductions of approximately 3.497 MW in active power and 14.422 MVAr in reactive power. In terms of system losses, the total active and reactive power losses are reduced by 6.96% and 0.74%, respectively, compared to the base case.

Overall, the findings confirm that the strategic deployment of UPFC devices provides an efficient and practical solution for enhancing power system stability, improving voltage regulation, and reducing transmission losses. The proposed methodology can be extended to other large-scale power systems to support optimal planning and operation.

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